

BRAC University

Department of Computer Science and Engineering

CSE422 — Artificial Intelligence

Tree Sterility Prediction

Dataset: Tree Sterility Dataset

Models: Logistic Regression | Gaussian Naive Bayes | Neural Network

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1. Introduction

This report presents a machine learning study completed for the CSE422 Artificial Intelligence Laboratory course at BRAC University. The main goal of the study is to predict tree sterility by identifying whether a seedling is likely to survive or die. The prediction is based on ecological and biochemical data collected from different forest plots.

Tree mortality affects biodiversity, carbon storage, and forest health. Predicting which seedlings may die can help protect forests more effectively. Traditional methods rely on expert knowledge and manual observation. However, ecological data is now large and complex so machine learning provides a faster and more practical solution.

The dataset used is the Tree Sterility Dataset, which captures light intensity, mycorrhizal associations, soil conditions, biochemical markers (Phenolics, Lignin, NSC) and census information across multiple forest plots. The target variable Event is binary: 1 = mortality event (died), 0 = survived.

The project includes the complete machine learning process, starting from data exploration and pre-processing. It uses supervised learning models such as Logistic Regression, Gaussian Naive Bayes and a Neural Network to classify the data. It also applies K-Means clustering for unsupervised learning and compares the performance of all models. Among them the Neural Network performed the best which achieved 92.8% accuracy and an AUC score of 0.985.

2. Dataset Description

2.1 Overview

The dataset was loaded from a CSV file and initially contained 2,783 rows and 24 columns. Each row represents a single tree seedling observation.

Feature	Type	Description
No	Integer	Unique seedling identifier
Plot / Subplot	Int / Category	Spatial plot location
Species	Categorical	Tree species (Acer saccharum, Quercus alba, etc.)
Light_ISF	Float	Indirect Site Factor — continuous light intensity
Light_Cat	Categorical	Categorized light level: Low / Med / High
Core	Integer	Year of core sample (2016 or 2017)
Soil	Categorical	Soil type / dominant adult tree species
Adult	Categorical	Adult tree identifier in the plot

Feature	Type	Description
Sterile / Conspecific	Categorical	Experimental treatment conditions
Myco / SoilMyco	Categorical	Mycorrhizal inoculation treatment
PlantDate	Categorical	Date of planting
AMF	Float	Arbuscular Mycorrhizal Fungi colonization (%)
EMF	Float	Ectomycorrhizal Fungi colonization — 54% missing
Phenolics	Float	Leaf phenolic content (biochemical marker)
Lignin	Float	Leaf lignin content (biochemical marker)
NSC	Float	Non-Structural Carbohydrates
Census	Integer	Number of census observations
Time	Float	Days since planting at last census
Event	Binary Float	TARGET: 1 = died, 0 = survived
Harvest / Alive	Categorical	Over 74% missing — dropped in pre-processing

2.2 Problem Type

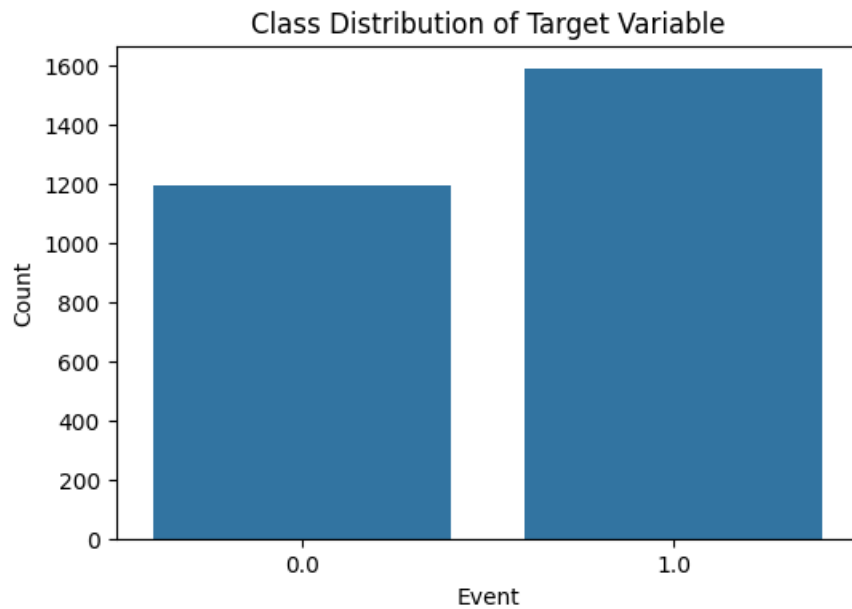
This is a binary classification problem. The output feature Event takes the value 1 (seedling died/sterile event) or 0 (survived). Classification is appropriate because we predict a discrete category rather than a continuous quantity.

2.3 Class Distribution — Imbalanced Dataset Check

After removing the single row with a missing target value (2,782 usable samples), the class distribution is:

Class Label	Count	Proportion
1.0 Event — Died	1,587	57.1%
0.0 No Event — Survived	1,195	42.9%

The dataset is mildly imbalanced — Class 1 has ~33% more instances than Class 0. This was accounted for during evaluation by reporting Precision, Recall, F1-Score, and AUC alongside accuracy.



2.4 Feature Types & Encoding Rationale

The 24 columns comprise 8 float64, 4 int64, and 12 object (categorical) features. Categorical features required encoding. Label Encoding was applied to all 10 categorical input features because the primary goal was predictive performance, and the models used (LR, NB, MLP) handle integer-coded categories well after standardization.

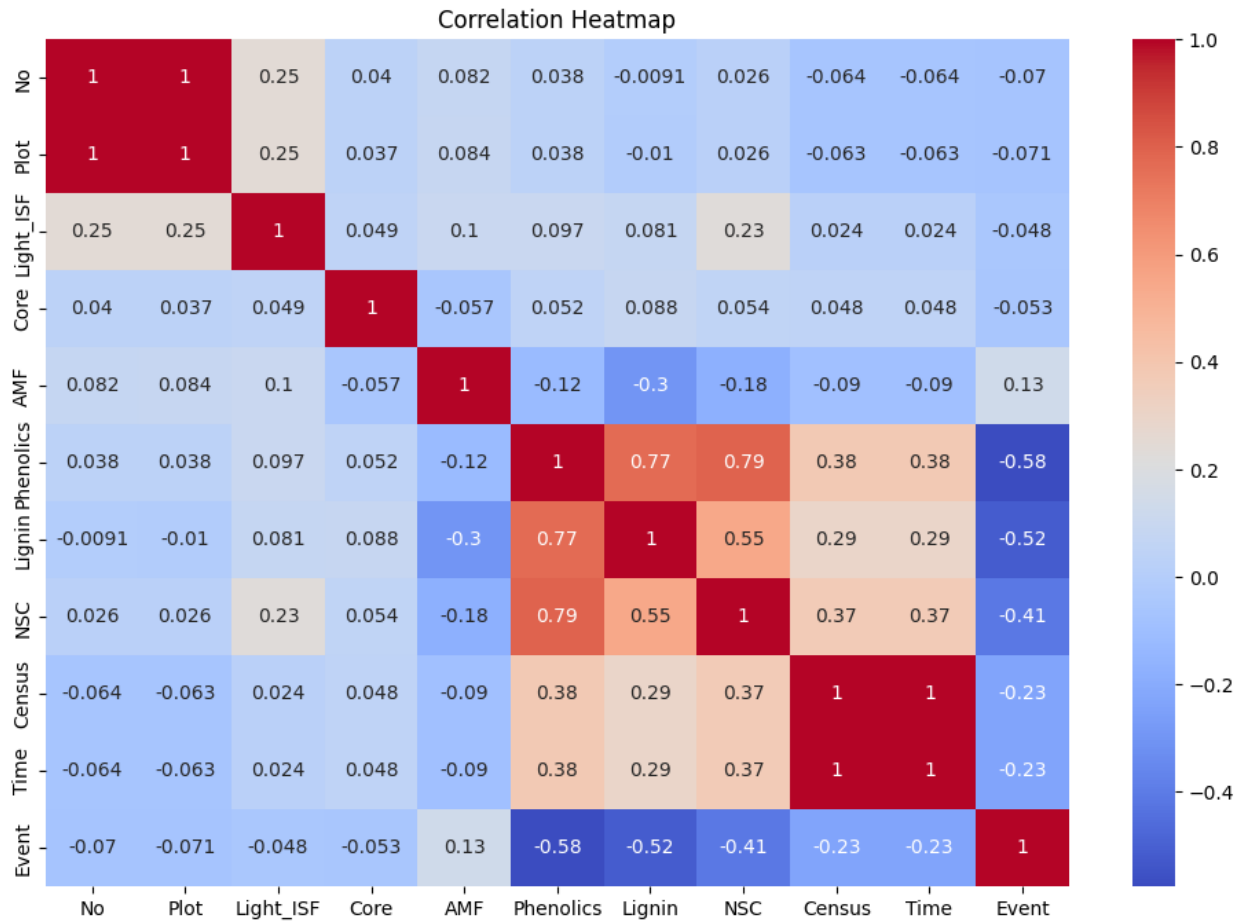
Correlation analysis was performed across all numerical features using a Pearson heatmap (Section 3) to understand linear relationships between input and output features before modelling.

3. Exploratory Data Analysis (EDA)

3.1 Correlation Heatmap

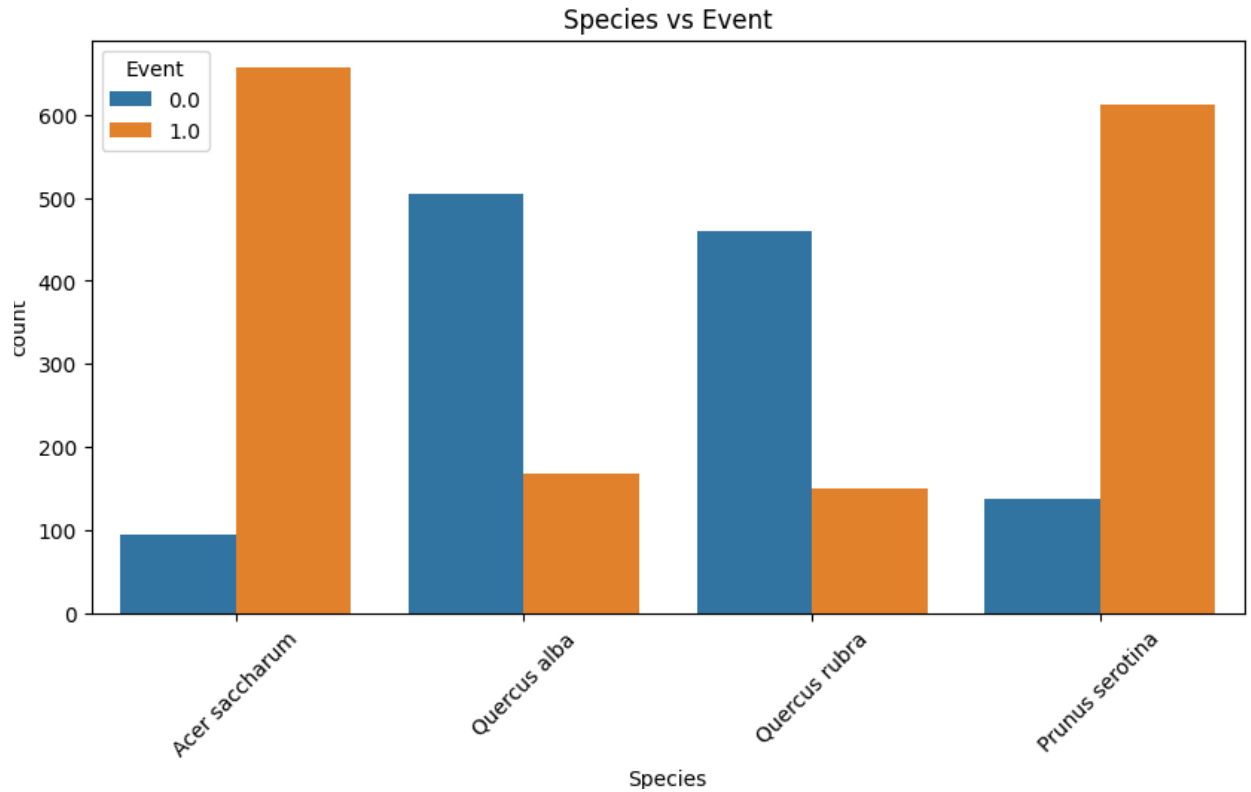
A Pearson correlation heatmap was generated for all numerical features. Key observations:

- **Time** and **Census** are strongly correlated ($r \approx 0.97$) — highly redundant features.
- **Lignin** and **NSC** show moderate positive correlation.
- **Event** shows a weak negative correlation with Time and Census — seedlings that died tended to do so early.
- **Phenolics** shows a weak positive correlation with Event.
- **AMF** shows low linear correlation with most features — its effect may be nonlinear.



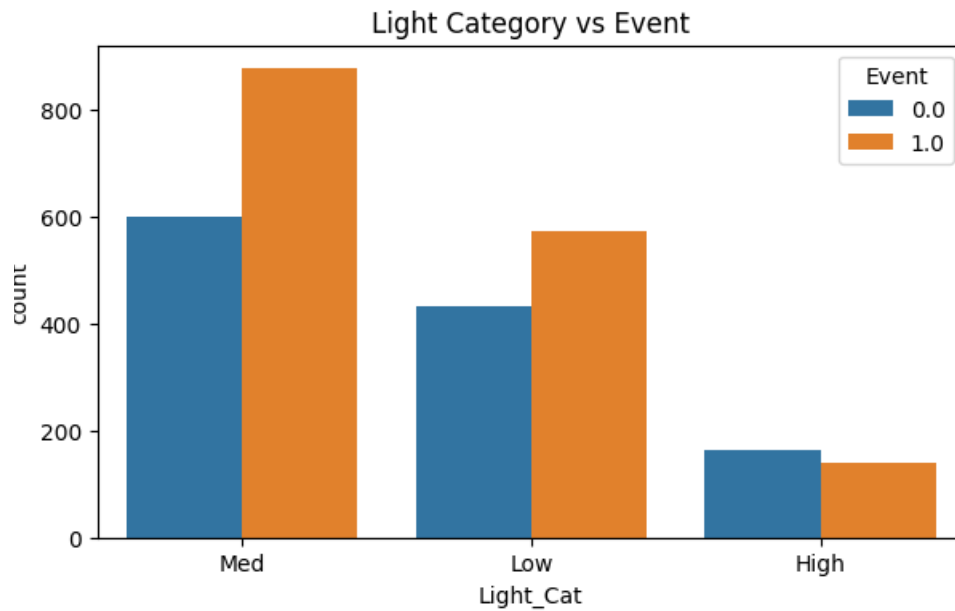
3.2 Species vs. Event

Acer saccharum (Sugar Maple) dominates the dataset and shows the highest absolute count of mortality events. Oak species (*Quercus alba*, *Quercus rubra*) show comparatively better survival rates, consistent with ecological literature on species-specific light sensitivity.



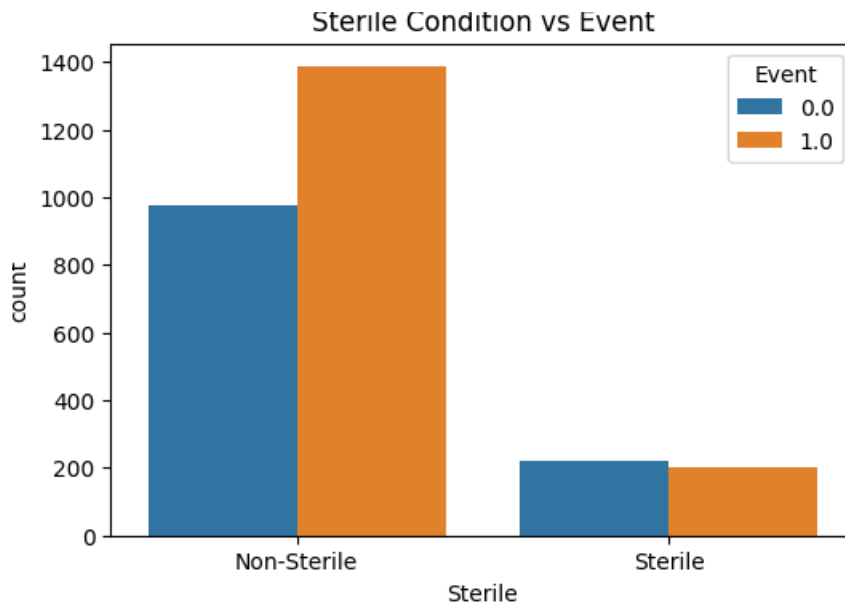
3.3 Light Category vs. Event

Seedlings in Low light conditions have a proportionally higher mortality rate, while those in High light environments survive more often. This confirms light as a primary driver of seedling success — a well-established finding in forest ecology.



3.4 Sterile Condition vs. Event

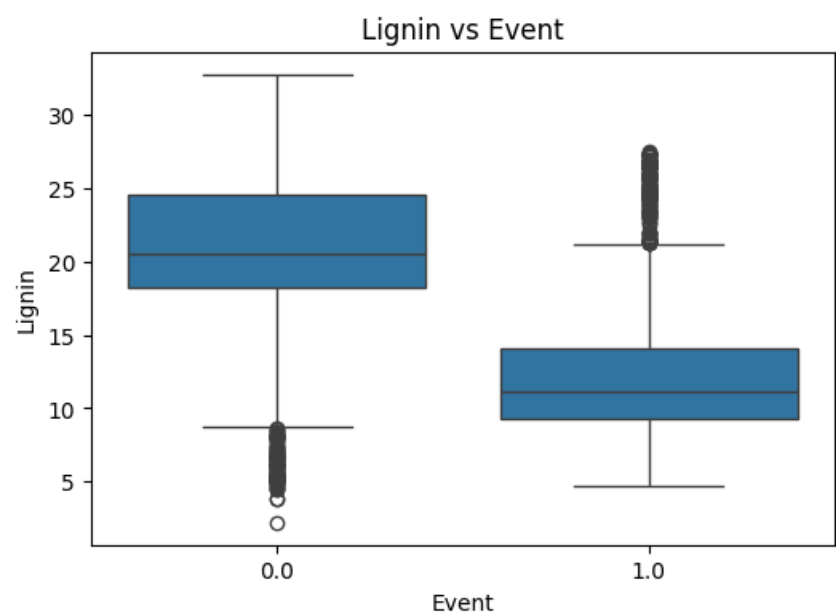
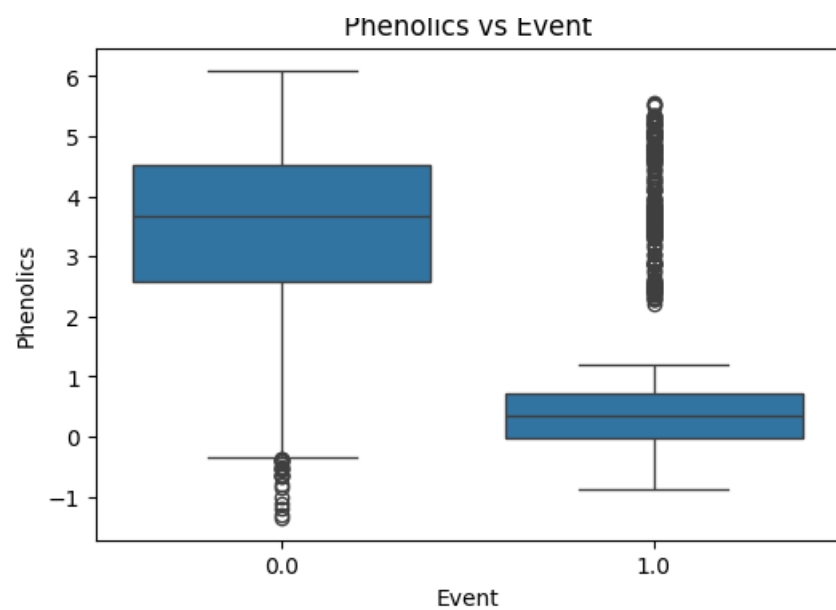
Seedlings planted in sterilized soil (mycorrhizal network removed) exhibit markedly higher mortality, reinforcing the understanding that mycorrhizal associations are critical for seedling survival. This experimental variable is one of the strongest categorical predictors.

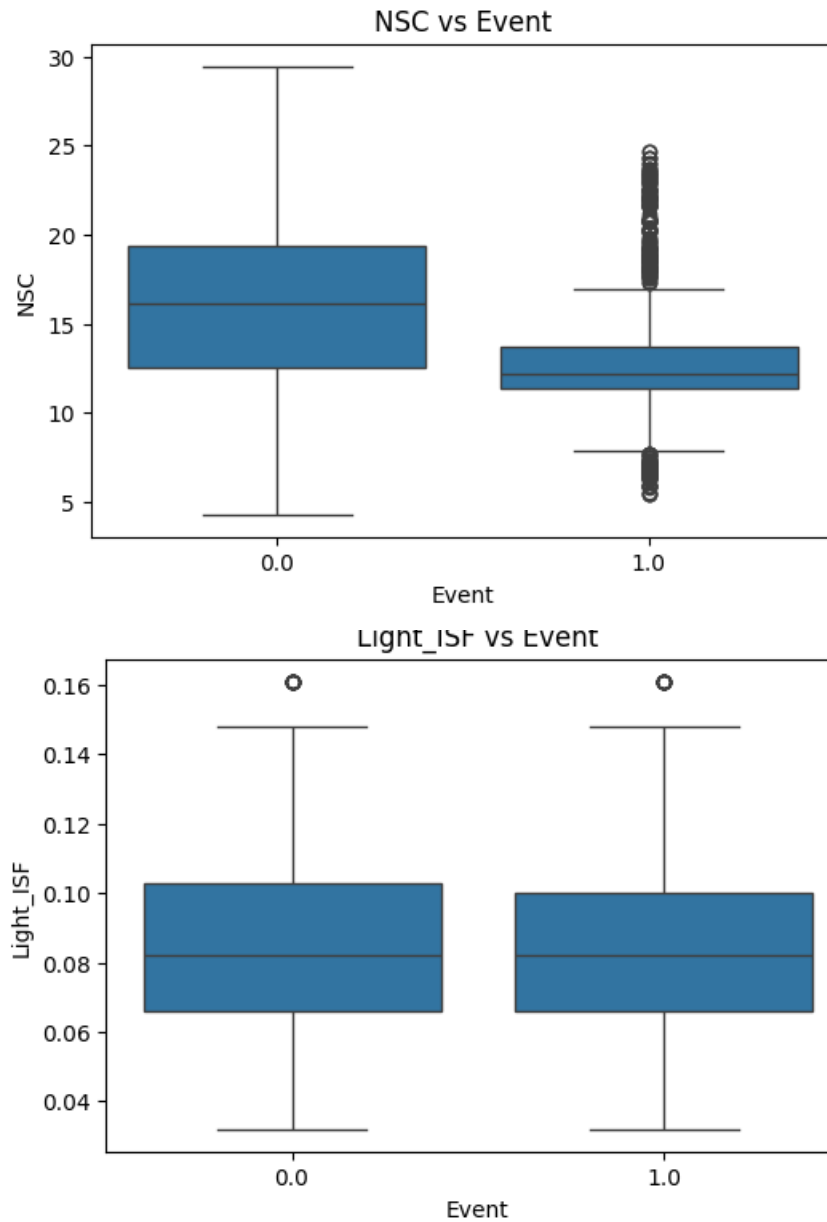


3.5 Biochemical Features vs. Event (Boxplots)

Boxplots for Phenolics, Lignin, NSC, and Light_ISF grouped by Event class reveal the following:

- **Phenolics:** Died seedlings (Event=1) show slightly higher median values and greater spread.
- **Lignin:** Survivors tend toward slightly higher Lignin — indicating more robust cell walls.
- **NSC:** Survivors show marginally higher non-structural carbohydrate reserves, consistent with carbon-starvation mortality theory.
- **Light_ISF:** Survivors receive slightly more light on average.





4. Dataset Pre-Processing

4.1 Faults Identified

Three categories of data quality problems were identified:

- **Null / Missing Values:** Three columns had critical missing rates; one target row was missing.
- **Categorical Values:** Ten string-type features required encoding.
- **Feature Scaling:** Numerical features span very different ranges, requiring standardization.

Column	Missing Values	Missing %	Action Taken
Alive	2,292	82.4%	Column dropped
Harvest	2,079	74.7%	Column dropped
EMF	1,500	53.9%	Column dropped
Event	1	0.04%	Row dropped
All others	0	0%	No action needed

4.2 Solutions Applied

- ❑ **Dropping Columns:** The columns *Alive*, *Harvest*, and *EMF* were removed because they contained too many missing values. Filling in more than half of the data could create bias and reduce model accuracy. After removing them, the dataset had 20 input features and 1 target column.
- ❑ **Handling Missing Values:** The remaining missing values were filled using Simple Imputer. Numerical data used the median value, while categorical data used the most common value. After this step, there were no missing values left.
- ❑ **Label Encoding:** The categorical columns were converted into numerical values using Label Encoder so the machine learning models could process them properly.
- ❑ **Feature Scaling:** StandardScaler was applied to all features to normalize the data. This helped improve the performance of Logistic Regression and the Neural Network.

Step	Technique	Reason
1. Drop columns	Remove Alive, Harvest, EMF	>50% missing — imputation would bias data
2. Drop row	Remove 1 row with missing Event	Target value cannot be imputed
3. Impute numerical	Median imputation	Robust to outliers
4. Impute categorical	Mode imputation	Preserves most common category
5. Encode	Label Encoding	Converts strings to integers for ML models
6. Scale	StandardScaler (z-score)	Equalizes feature ranges; required for LR & MLP

5. Dataset Splitting

The processed dataset (2,782 samples, 20 features) was split using stratified random splitting with the stratify=y parameter, preserving the ~57%/43% class ratio in both subsets. An 80/20 split was used.

Split	Samples	Proportion	Class 0 (approx.)	Class 1 (approx.)
Training Set	2,225	80%	956	1,269
Test Set	557	20%	239	318

An 80/20 split provides sufficient training data while maintaining a meaningful held-out test set for unbiased evaluation. Stratification ensures the mild class imbalance is represented equally in both splits.

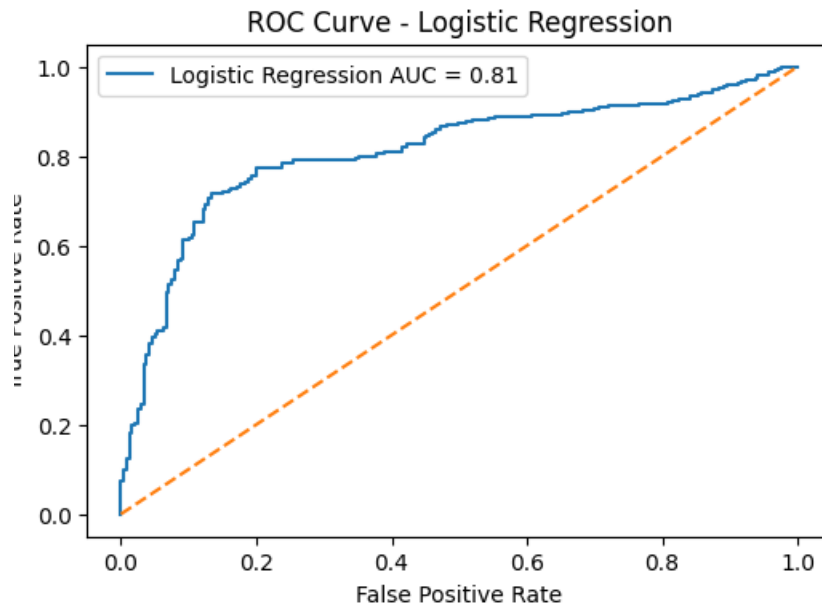
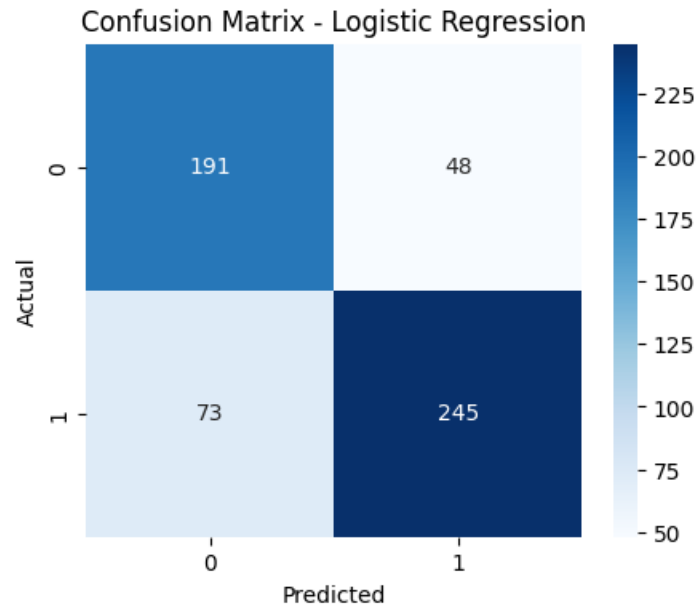
6. Model Training & Testing (Supervised)

6.1 Logistic Regression

Logistic Regression is a machine learning model used for classification. It predicts the probability of a seedling surviving or dying based on the input features. In this study, the model was trained with max_iter=1000 to make sure the training process completed properly. It was used as the baseline linear model for comparison.

Results: Logistic Regression achieved 78.28% accuracy. The model showed good precision (83.62%), but the recall was lower (77.04%). This means it missed around 23% of the actual seedling deaths. The lower performance is likely because many ecological patterns are complex and not fully captured by a linear model.

Accuracy	Precision	Recall	F1 Score	AUC Score
78.28%	83.62%	77.04%	80.20%	81.10%

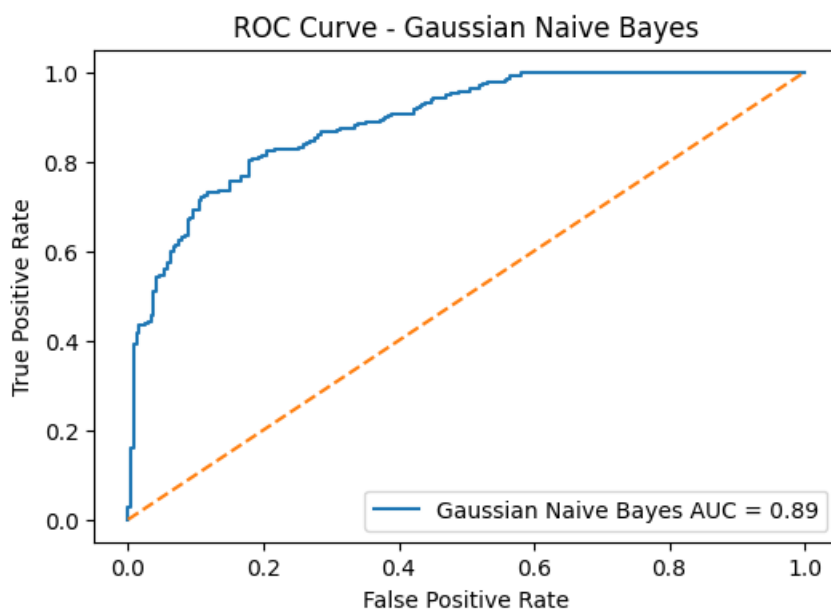
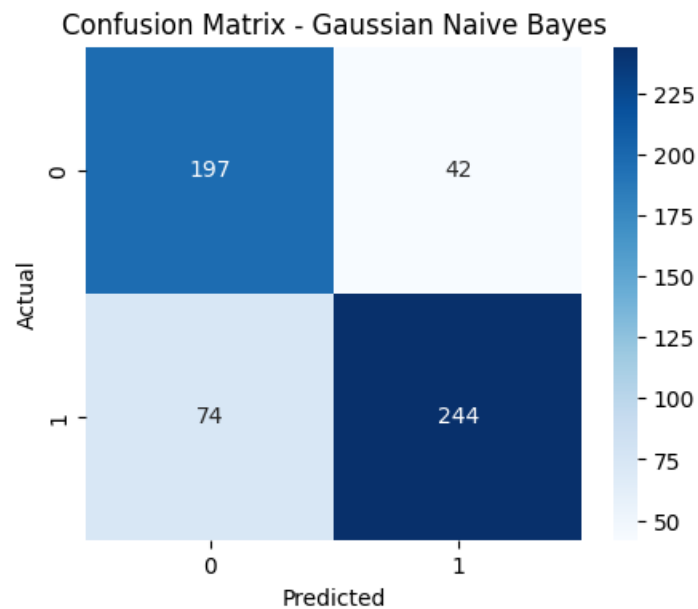


6.2 Gaussian Naive Bayes

Gaussian Naive Bayes is a classification model that assumes the features are independent of each other for each class. It treats each feature as following a normal (Gaussian) distribution. Even with this simple assumption, the model can still perform well when the data is scaled and close to a normal distribution.

Results: Naive Bayes performed slightly better than Logistic Regression, achieving 79.17% accuracy. It also achieved a higher AUC score of 0.8897, showing that the model was good at predicting probabilities and separating the classes across different decision thresholds.

Accuracy	Precision	Recall	F1 Score	AUC Score
79.17%	85.31%	76.73%	80.79%	88.97%



6.3 Neural Network — MLP Classifier

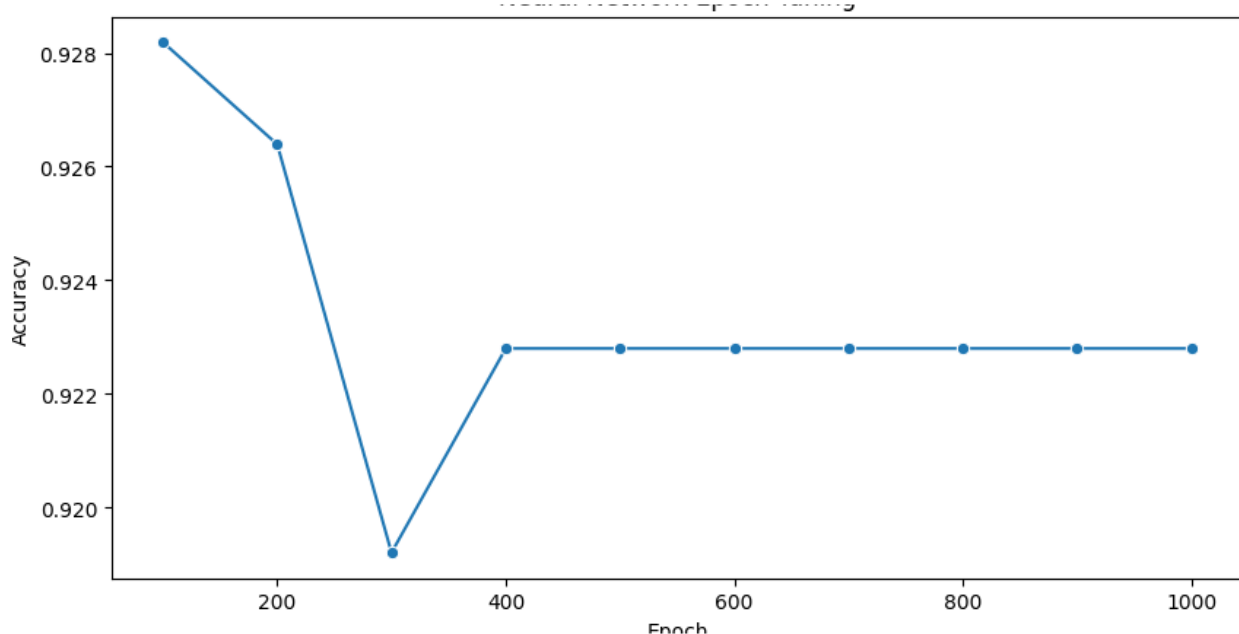
A Multi-Layer Perceptron (MLP) was built using scikit-learn's `MLPClassifier`. The model has two hidden layers with 64 and 32 neurons. It uses the ReLU activation function and the Adam optimizer. This structure helps the model learn complex and nonlinear patterns in the ecological data.

6.3.1 Epoch (max_iter) Tuning

Rather than arbitrarily selecting a fixed epoch count, a systematic sweep was conducted across [100, 200, 300, ..., 1000] epochs, evaluating accuracy at each step. This ensures the optimal number of training iterations is used.

Epoch	Accuracy	Precision	Recall	F1 Score
100	92.82%	94.84%	92.45%	93.63%
200	92.64%	94.82%	92.14%	93.46%
300	91.92%	94.46%	91.20%	92.80%
400	92.28%	94.21%	92.14%	93.16%
500–1000	92.28%	94.21%	92.14%	93.16%

The best accuracy was achieved at epoch 100 (92.82%), after which performance plateaued at ~92.28% for epochs 400–1000. This rapid convergence is typical of well-scaled datasets with the Adam optimizer's adaptive learning rates.



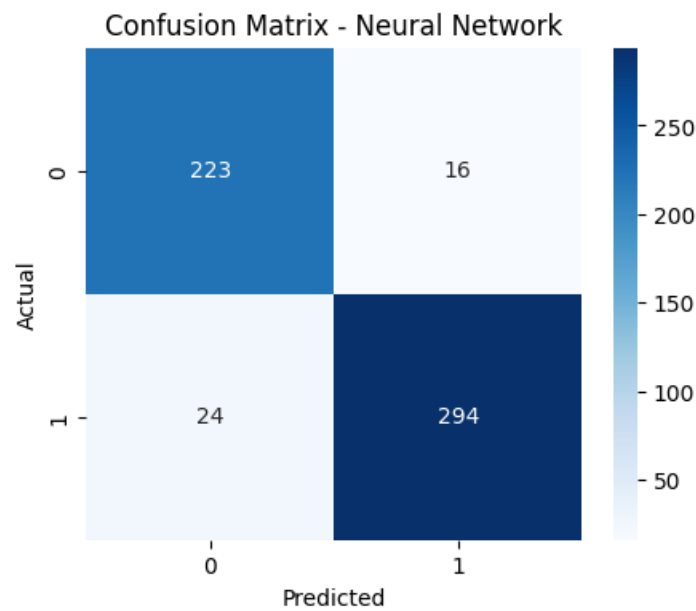
6.3.2 Final Neural Network Results

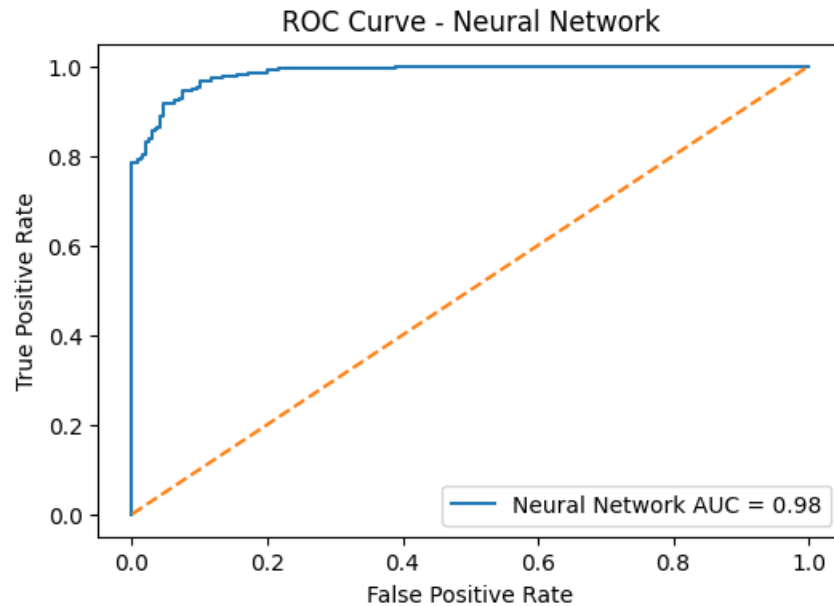
The final MLP (epoch=100) dramatically outperformed both linear models, achieving outstanding scores across all five metrics:

Accuracy	Precision	Recall	F1 Score	AUC Score
92.82%	94.84%	92.45%	93.63%	98.50%

The AUC of 0.985 is particularly noteworthy — it indicates the Neural Network can almost perfectly rank all seedlings by their true mortality risk, regardless of the classification threshold used.

Class	Precision	Recall	F1 Score	Support
0.0 (Survived)	90%	93%	92%	239
1.0 (Died)	95%	92%	94%	318





7. Unsupervised Learning — K-Means Clustering

In addition to supervised classification, the dataset was treated as an unsupervised learning problem using K-Means clustering — exploring natural groupings in the data without using the Event label.

7.1 Feature Selection & Configuration

Four biochemical and environmental features were selected for clustering: Phenolics, Lignin, NSC, and Light_ISF. These represent key physiological indicators of seedling health and resource availability. All features were standardized before clustering.

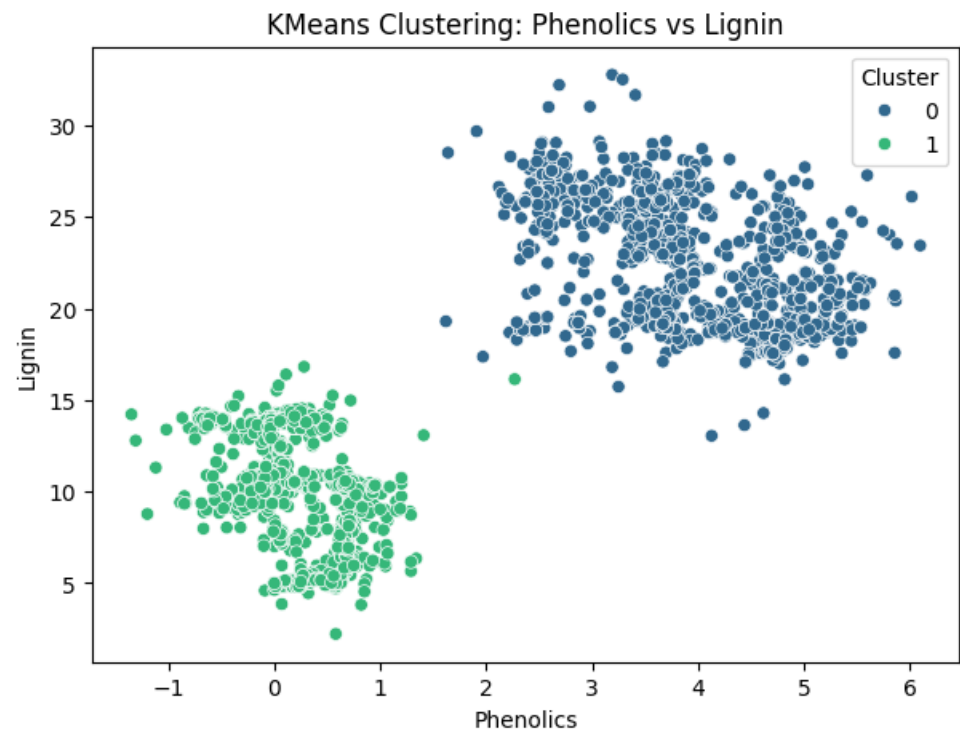
K-Means was configured with $k=2$ clusters (mirroring the binary Event variable) and $n_init=10$ random initializations to minimize sensitivity to starting conditions.

7.2 Results

Cluster	Count	Proportion
Cluster 0	1,281	46.1%
Cluster 1	1,501	53.9%

The cluster sizes (53.9% and 46.1%) are very close to the real class distribution (57.1% and 42.9%). This shows that even without using labels, the biochemical features still contain useful information about which seedlings belong to different survival groups.

The scatter plot shows two groups that are partly separate but also overlap when plotted using Phenolics and Lignin. This overlap explains why biochemical features alone cannot fully separate the classes. Supervised models perform better because they also use other information like species, light, and soil conditions, which improves accuracy.



8. Model Selection & Comparison Analysis

8.1 Summary Comparison Table

All three supervised models were evaluated on the same held-out test set (557 samples) using five standard metrics.

Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	78.28%	83.62%	77.04%	80.20%	81.10%
Gaussian Naive Bayes	79.17%	85.31%	76.73%	80.79%	88.97%
Neural Network (MLP)	92.82%	94.84%	92.45%	93.63%	98.50%

8.2 Accuracy Comparison

The bar chart shows about a 14% difference between the Neural Network (92.82%) and the linear models (around 78–79%). In real ecological use, this means the Neural Network would

miss far fewer seedling deaths. This is important for better conservation planning and decision-making.

8.3 Precision, Recall & F1 Score Comparison

The Neural Network gives the best overall performance, with an F1 score of 93.63%. The linear models have higher precision than recall, meaning they are more careful when predicting death. They usually predict survival unless they are very sure about a death case. Because of this, they miss more actual deaths (false negatives).

8.4 AUC Score Comparison.

AUC (Area Under the ROC Curve) measures discriminative ability across all thresholds. The Neural Network's AUC of 0.985 is exceptional — a random classifier scores 0.5 and a perfect classifier scores 1.0. Gaussian Naive Bayes achieves a respectable 0.890, while Logistic Regression trails at 0.811.

8.5 Why the Neural Network Outperforms

- **Non-linearity:** ReLU activations allow the MLP to capture complex interactions (e.g., the combined effect of low light AND sterilized soil) that linear models cannot model.
- **Automatic feature interaction:** The two hidden layers (64→32) discover cross-feature patterns without manual feature engineering.
- **Efficient optimization:** Adam automatically adjusts learning rates for faster, more efficient learning. In this dataset with 20 features, it reaches near-optimal performance by about the 100th epoch.
- **Sufficient data:** With 2,225 training samples, the MLP learns nonlinear patterns without overfitting — smaller datasets would favour simpler models.

9. Conclusion

9.1 Summary of Findings

- The dataset is a binary classification problem with mild class imbalance (57% positive). Pre-processing removed 3 high-missing columns, imputed remaining gaps, encoded 10 categorical features, and standardized all inputs.
- EDA revealed that light availability, mycorrhizal treatment, species, and biochemical markers (Phenolics, Lignin, NSC) are the strongest predictors of mortality.
- **Logistic Regression** achieved 78.28% accuracy — a solid linear baseline, limited by ecological non-linearity.
- **Gaussian Naive Bayes** achieved 79.17% accuracy and a notably higher AUC of 0.890, showing strong probabilistic ranking despite the independence assumption.
- The **Neural Network (MLP)** outperformed both with 92.82% accuracy and AUC = 0.985. The 64→32 architecture with epoch=100 was optimal.

- **K-Means clustering** on biochemical features produced two clusters closely mirroring the actual class distribution, validating the ecological interpretability of the chosen features.

9.3 Challenges Faced

- **High missing data:** Three columns had >50% missing values, forcing their removal. EMF (ectomycorrhizal fungi) is ecologically significant — losing this feature likely costs some predictive power.
 - **Label Encoding ordinality:** Assigning integer codes to species or soil types implies a false ordinal relationship. One-Hot Encoding would be more statistically rigorous but would greatly expand the feature space.
 - **Mild class imbalance:** The 57/43 split may bias models toward the majority class. SMOTE oversampling or class-weight adjustments could improve minority-class recall.
 - **Interpretability:** The Neural Network's superiority comes at the cost of explainability. Ecologists need to understand why a seedling is predicted to die — SHAP values or a Random Forest's feature importances could address this.
-